

Intelligent Traffic Management System (ITMS)

Real-Time Computer Vision, Reinforcement Learning Adaptive Signal Control, and 30-Second Police Dispatch SLA

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Repository:	github.com/NAVYA1174/intelligent-traffic-management-system	Tech Stack:	Python, YOLOv8, PyTorch, RL (DQN), FastAPI

PROJECT ABSTRACT

Urban congestion and traffic gridlock impose significant economic losses, environmental pollution, and public safety hazards in modern metropolitan areas. Conventional fixed-time traffic light systems operate on rigid schedules unable to adapt to real-time traffic surges, leading to prolonged delays, increased fuel consumption, and slow emergency responses. This project presents the Intelligent Traffic Management System (ITMS), an end-to-end cyber-physical platform combining real-time computer vision and reinforcement learning to dynamically optimize urban traffic flows and automate emergency incident handling.

The system processes high-definition CCTV camera streams using a custom YOLOv8 object detection and multi-object tracking pipeline. It accurately detects, counts, and classifies vehicles into passenger cars, transit buses, heavy freight trucks, motorcycles, and emergency vehicles, estimating approach speeds and lane-level queues. A restricted polygon zone tracker detects illegal parking violations in tow-away corridors, while kinetic trajectory analysis identifies sudden collisions and stalled vehicles. To mitigate congestion, an adaptive Deep Q-Network (DQN) and Q-learning agent continuously reads multi-approach queue states and dynamically modulates signal phase durations, prioritizing dense approaches and preempting green-wave corridors for emergency ambulances. Benchmark simulations demonstrate that ITMS reduces average commuter waiting times by 45% to 58% compared to traditional fixed-time controllers, concurrently yielding substantial reductions in vehicular idle emissions and fuel consumption. Furthermore, ITMS integrates a rapid Incident Detection and Alert System enforcing a strict 30-second notification SLA for municipal traffic police departments, capturing timestamped evidence snapshots and enabling immediate patrol dispatch. An interactive Traffic Operations Center (TOC) web dashboard delivers city-wide GIS mapping, live annotated video streams, active signal head countdowns, and comprehensive telemetry.

1. PROBLEM STATEMENT & OBJECTIVES

Contemporary metropolitan intersections rely overwhelmingly on fixed-cycle electro-mechanical signal heads. During peak hours, asymmetric directional demand leaves empty corridors green while congested lanes remain stalled at red lights. Furthermore, urban traffic accidents and illegal parking in transit lanes frequently go undetected for 10–25 minutes until reported by manual citizen calls, severely delaying police, ambulance, and towing dispatch. The objectives of ITMS are:

- **Real-time Computer Vision:** Detect and classify 5 vehicle classes with sub-40ms latency using YOLOv8.
- **Adaptive Traffic Control:** Implement an RL agent (DQN / Q-Learning) reducing average intersection delay by >40%.
- **Emergency Vehicle Preemption:** Detect sirens/ambulances and clear green waves without manual intervention.
- **30-Second Incident SLA:** Detect collisions and tow-away parking, alerting traffic police within 30 seconds.
- **Unified Operations Dashboard:** Provide an interactive GIS map, live camera feeds, and automated police dispatch.

2. SYSTEM ARCHITECTURE & METHODOLOGY

ITMS comprises four tightly coupled subsystems: Computer Vision Sensing, Reinforcement Learning Controller, Incident SLA Engine, and the Web Operations Center Dashboard.

2.1 Computer Vision & Vehicle Tracking (YOLOv8)

The computer vision engine leverages a customized YOLOv8 nano model trained on COCO vehicle classes. Detections are paired with an adaptive Euclidean centroid tracker maintaining trajectory history and stationary duration counters. Instantaneous velocity is computed via perspective-calibrated pixel displacement. Illegal parking is flagged when vehicle centroids dwell inside polygonal Tow-Away zones beyond a grace threshold, while accidents are detected through abrupt velocity drops paired with high bounding box IoU overlap.

2.2 Reinforcement Learning Adaptive Signal Control

Unlike rigid timers, the ITMS controller frames traffic signal optimization as a Markov Decision Process (MDP). The state vector includes queue lengths across North, South, East, and West corridors, current green phase, elapsed green time, and emergency preemption flags. The action space permits phase extensions or safe transitions through yellow intervals. The reward formulation balances queue reduction, cumulative delay penalties, and emergency clearance bonuses:

Reward: $R_t = -0.5 \sum Q_{\text{waiting}} - 0.2 \sum Q_{\text{active}} + 2.0 \times \text{Discharged} + 15.0 \times \text{Emergency_Cleared}$

2.3 Incident Detection & 30-Second Police Dispatch SLA

The incident engine monitors collisions, vehicle stalls, and illegal parking. Once identified, an incident record is logged with a 30-second target SLA timer. High-resolution evidence frames with bounding-box overlays and timestamps are recorded in the evidence repository. Operations personnel can dispatch designated patrol units with a single click, recording officer timestamps for municipal accountability.

3. EXPERIMENTAL BENCHMARK RESULTS

The system was evaluated across four high-density urban corridors (Downtown Central, Tech Corridor, Metro Ring Road, Harbor Freight) under asymmetric and surging traffic conditions. Performance was compared directly against a standard 40-second fixed-time baseline controller:

Performance Metric	Fixed-Time Baseline	ITMS (YOLO + RL DQN)	Relative Improvement
Average Commuter Delay	28.6 seconds / veh	12.4 seconds / veh	-56.6% (Delay Reduction)
Peak Intersection Queue	18 vehicles / approach	7 vehicles / approach	-61.1% (Queue Reduction)
Vehicular Throughput	1,420 veh / hour	1,940 veh / hour	+36.6% (Throughput Gain)
Emergency Clearance Time	44.2 seconds	8.1 seconds	-81.6% (Fast Clearance)
Incident Alert Dispatch SLA	14.5 minutes (Manual)	< 30 seconds (Automated)	96.5% Faster Response
Estimated Idle CO2 Emissions	Baseline Benchmark	-4.82 kg CO2 / hr / node	Significant Carbon Cut

4. CONCLUSION & MUNICIPAL DEPLOYMENT

The Intelligent Traffic Management System demonstrates the practical efficacy of uniting real-time deep learning computer vision with reinforcement learning for modern urban infrastructure. By replacing static time-clocks with responsive, data-driven controllers, municipalities can alleviate recurring bottlenecks, reduce carbon footprints, and dramatically expedite emergency response times. The complete source code, trained neural network weights, and operations dashboard are available in the public repository.